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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV Sem B.Tech (CBCS) Mid Semester Examinations - II, April 2025

MA2005-1 – LINEAR ALGEBRA AND ITS APPLICATIONS

(For AM, CB, CC, CS, IS, RI)

Duration: 1 Hour

Max. Marks: 20

Part – A: Multiple Choice Questions

*Note: Answer all **Four** questions in the **Answer Book**. Each question carries equal marks.*

1. **In three dimensional Euclidean space inverse of (-2,-2,0) with respect to vector addition is**
   A) (2, 2, 2)
   B) (-1,-1,0)
   C) (-2,-2,0)
   D) (2, 2, 0)

2. **If a basis of a vector space has 3 elements, then dimension of the vector space is**
   A) 2
   B) 1
   C) 3
   D) 0

3. Let $T: R^3 \to R^4$ be the linear transformation defined by $T(x_1, x_2, x_3) = (x_1, 0, x_2, x_2)$. Then basis for null space of T is
   A) (1,0,0)
   B) (0,0,1)
   C) (0,1,0,0)
   D) (1,1,0,0)

4. Which of the following statement is not true if $T: U \to V$ is a linear transformation
   A) V is always the range of T
   B) Null space of T is a subspace of U
   C) Range space of T is a subspace of V
   D) $T(0)=0$

Part – B: Descriptive Answer Questions

*Note: Answer any **One** full question from **each Unit.***

			Marks	BT*	CO*	PO*
		Unit – I				
1.	a)	i) Define a subspace of a vector space V over a field F. ii) If G and H are subspaces of a vector space V over F then prove that $G \cap H$ is also a subspace of V.	4	L*1	3	1
	b)	Define (i) Row space (ii) Null space of a matrix. Check whether the vector $\begin{pmatrix} 3 \\ 3 \\ -4 \end{pmatrix}$ is in the column space of the matrix $A = \begin{pmatrix} 1 & -3 & 4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{pmatrix}$	4	L2	3	2
2.	a)	i) Define Linearly dependent and Linearly independent of a set of vectors. ii) Express $u=(1,-2,5)$ in R^3 as a linear combination of the vectors $u_1=(1,1,1)$ $u_2=(1,2,3)$ $u_3=(2,-1,1)$ (-6, 3, 2)	5	L1	3	2
	b)	Find the dimension of the subspace W of R^3 spanned by the vectors {(2,4,2), (1,-1,0),(1,2,1),(0,3,1)}. Extend the basis of W to a basis of R^3. [first 2 cols	3	L2	3	1
		Unit – II				
3.	a)	(i)State Rank-Nullity theorem. (ii) Let a linear transformation T: $R^3 \to R^3$ be defined by $T(x,y,z)=(x+2y-z, y+z, x+y-2z)$. Find the null space of T and range space of T. Also find dimension of the null space of T and dimension of range space of T.	5	L3	4	2

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b) Check whether a transformation T: $R^2 \to R^2$ defined by $T(x,y)=(x+y,x)$ is a linear transformation or not. 3 L2 4 1

4. a) Find the matrix of the linear transformation T: $R^3 \to R^2$ defined by $T(x,y,z)=(x-y+2z, 3x+y)$ with respect to bases $B=\{(1,1,1),(1,2,3),(1,0,0)\}$ and $B'=(1,1),(1,-1)$

$$\begin{bmatrix} 3 & -1 \\ 5 & 0 \\ 5 & 2 \end{bmatrix}$$

4 2

b) Let V and W be vector spaces over the field F. Let T and U be linear transformation from V into W. Then prove that the function (T+U) defined by (T+U)α=T(α)+U(α) is a linear transformation from V into W. Also prove , if c is any element of F, the function(cT) defined by (cT)(α)=c(Tα) is a linear transformation from V into W. 3 L2 4 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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T(1,1,1)

1-1+2 = 2
3+1 = 4

1,2,3
1-2+2(3) (5,5)
3+2

1,0,0
1-0+0 (1,3)

(2,3) = $a_{11}(1,1) + a_{21}(1,-1)$

$a_{11} + a_{21} = 2$ 5 1

$a_{11} - a_{21} = 4$ 5 3

$a_{11} + a_{11} + 1 = 2$

$6a_{11} = 6$ 3

$a_{21} = -1$

$$\begin{bmatrix} 3 & 5 & 2 \\ -1 & 0 & -1 \end{bmatrix}$$

a 5

2a

a_{11} a_{11}

a_{21}

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Off-Campus Centre of Nitte (Deemed to be University)

IV Sem B.Tech (CBCS) Mid Semester Examinations-II, April 2025

CS2103-1 – SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

(CS, CB)

Duration: 1 Hour

Max. Marks: 20

Part – A: Multiple Choice Questions (1*4 = 4 marks)

*Note: Answer all **Four** questions in the **Answer Book**. Each question carries equal marks.*

1. **Which of the following diagrams is used to model the vocabulary of a system?**
   A) Object Diagram
   B) Activity Diagram
   C) Class diagram
   D) Interaction Diagram
2. **Composition is a stronger form of which of the following?**
   A) Aggregation
   B) Inheritance
   C) Encapsulation
   D) Generalization
3. **Select the option that suits the Manifesto for Agile Software Development**
   A) Individuals and interactions
   B) Working software
   C) Customer collaboration
   D) All the above mentioned choices are correct
4. **What encapsulates both data and data manipulation functions?**
   A) Object
   B) Class
   C) Super Class
   D) Sub Class

Part – B: Descriptive Answer Questions (2*8 = 16 marks)

*Note: Answer any **One** full question from **each Unit**.*

			Marks	BT*	CO*	PO*
		Unit – I				
1.	a)	With a neat diagram explain Context model of MHC-PMS.	4	L2	3	2
	b)	Draw and explain the Sequence diagram for View Patient Information.	4	L2	3	2
2.	a)	Explain the Activity model of an Insulin's pump operation.	4	L2	3	2
	b)	Outline the Use case description for Report Weather.	4	L2	3	2
		Unit – II				
3.	a)	Explain various principles of Agile methods.	4	L2	3	2
	b)	Explain Test Driven Development Process with diagram.	4	L2	4	2
4.	a)	Explain any four Extreme Programming practices.	4	L2	3	2
	b)	Explain the Model of software testing process with diagram.	4	L2	4	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV Sem B.Tech (CSE) Mid Semester Examinations - II, APRIL 2025

CS3005-1 – MICRPROCESSOR AND EMBEDDED SYSTEM

Duration: 1 Hour | Max. Marks: 20

Part – A: Multiple Choice Questions (1*4 = 4 marks)

*Note: Answer all **Four** questions in the **Answer Book**. Each question carries equal marks.*

1. **Which of the following is NOT an advantage of modular programming?**
   A) Code reusability
   B) Easier debugging
   C) Increased execution time
   D) Better program organization
2. **Which interrupt is used to set video mode in 8086?**
   A) INT 10H, AH = 00H
   B) INT 16H, AH = 00H
   C) INT 21H, AH = 01H
   D) INT 13H
3. **Which register in ARM holds the flags indicating the status of the processor?**
   A) General-purpose register
   B) Program Counter (PC)
   C) Stack Pointer (SP)
   D) Program Status Register (PSR)
4. **What is the size of the general-purpose registers in ARM processors?**
   A) 8 bits
   B) 16 bits
   C) 32 bits
   D) 64 bits

Part – B: Descriptive Answer Questions (2*8 = 16 marks)

*Note: Answer any **One** full question from **each Unit**.*

		Marks	BT*	CO*	PO*
Unit – I					
1. a)	Compare and contrast RISC and CISC architectures	4	L2	3	1
b)	Write the procedure for reading a 2-digit hexa decimal number from the standard input device.	4	L3	3	1
2. a)	Discuss ARM design Philosophy.	4	L2	3	1
b)	Write an ALP to read an alphanumeric character and display its equivalent ASCII code at the center of the screen	4	L3	3	1
Unit – II					
3. a)	Write an embedded C program to control the speed of rotation of a stepper motor using potentiometer.	4	L3	4	1
b)	What is Arduino communication and explain the types of Arduino communication.	4	L2	4	1
4. a)	With a code snippet, explain pinMode(), digitalWrite(), and analogRead() functions.	4	L2	4	1
b)	Write an Embedded C program to scroll the word 'FIRE' from left to right on a 7-segment display.	4	L3	4	1

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Off-Campus Centre of Nitte (Deemed to be University)

IV SEM B.Tech (CBCS) Mid Semester Examinations - II, April 2025

CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour

Max. Marks: 20

Part – A: Multiple Choice Questions (1*4 = 4 marks)

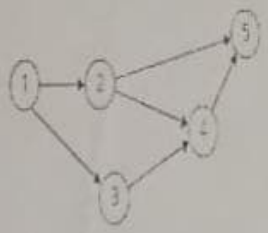
*Note: Answer all **Four** questions in the **Answer Book**. Each question carries equal marks.*

1. **Which of the following is a necessary condition for a graph to have a topological sorting?**
   A) The graph must be a tree
   B) The graph must be a DAG
   C) The graph must be a complete graph
   D) The graph must be a bipartite graph
2. **What type of data structure is used in Depth-First Search (DFS)?**
   A) Queue
   B) Heap
   C) Priority queue
   D) Stack
3. **What is the time complexity of distribution counting sort?**
   A) $\Theta(n^2)$
   B) $O(n\log n)$
   C) $O(2^n)$
   D) $\Theta(n)$
4. **In dynamic programming, the solution to a problem is built up from:**
   A) A series of overlapping sub problems
   B) A single recursive call
   C) A greedy approach
   D) A divide and conquer method

Part – B: Descriptive Answer Questions (2*8 = 16 marks)

*Note: Answer any **One** full question from **each Unit**.*

		Marks	BT*	CO*	PO*
Unit – I					
1. a)	Outline Insertion sort algorithm and show the sorting of list E, X, A, M, P, L, E in alphabetical order using insertion sort method.	4	L3	3	2
b)	Apply heap sort algorithm to sort the list 90, 89, 56, 72, 67, 12, 21, 34, 45.	4	L3	3	2
2. a)	Define AVL tree and construct AVL tree for the data {2, 9, 3, 12, 15, 8, 11}	4	L3	3	2
b)	Define topological sorting. Illustrate the topological sorting using Source Removal method for the following graph.	4	L3	3	2



Unit – II					
3. a)	Write an algorithm to sort list of integers using Comparison Counting and compare its time complexity with the time complexity of selection sort and Bubble sort	4	L3	4	2
b)	Summarize the steps of Horspool string matching algorithm and search for the pattern TIME in the text A STITCH IN TIME SAVES NINE	4	L3	4	2

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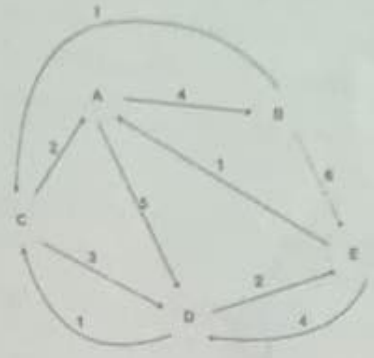
4. a) Apply All Pair Shortest path algorithm for the following graph.



b) Write an algorithm for Breadth-First Search (BFS). List any two facts that differentiate BFS from DFS. 4 L3 4 2

4 L3 3 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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IV Sem B.Tech (CBCS) Mid Semester Examinations - II, APRIL 2025

CS2102-1 – DATABASE MANAGEMENT SYSTEMS

CS, IS, AM, AD,CC

Duration: 1 Hour | Max. Marks: 15

Part – A: Multiple Choice Questions (1*3 = 3 marks)

*Note: Answer all **Three** questions in the **Answer Book**. Each question carries equal marks.*

1. **What is the result of the query SELECT COUNT (DISTINCT Salary) FROM EMPLOYEE;?**
   A) Total number of employees
   B) Total number of salaries greater than a specified value
   C) Total number of unique salaries
   D) Average salary of employees
2. **What type of dependency does 2NF aim to eliminate?**
   A) Partial dependency
   B) Referential dependency
   C) Transitive dependency
   D) Candidate key dependency
3. **In a relation, if attribute B is functionally dependent on attribute A, which of the following is true?**
   A) Each value of B is associated with one value of A
   A → B
   B) Each value of A is associated with one value of B
   C) There is no dependency between A and B
   D) A and B must be the same attribute

Part – B: Descriptive Answer Questions (2*6 = 12 marks)

*Note: Answer any **One** full question from **each Unit**.*

Unit – I

		Marks	BT*	CO*	PO*
1.	a) Consider the following schema and write the SQL queries for the following: ***Suppliers*** (**sid**: integer, **sname**: string, **address**: string) ***Parts*** (**pid**: integer, **pname**: string, **color**: string) ***Catalog*** (**sid**: integer, **pid**: integer, **cost**: double) i. Find the *sids* of suppliers who supply some red or green part ii. Find the *sids* of suppliers who supply only red parts iii. Find the *pnames* of parts for which there is some supplier iv. Find the *sids* of suppliers who supply some red part or are located at 221 Packer Street. (**solution should be in Nested Query**)	6	L3	3	2
	a) Consider the following schema and write the SQL queries for the following: **Movie** (movie_id, title, release_year, genre, director_id, rating) **Director** (director_id, name, birth_year, nationality) **Actor** (actor_id, name, birth_year, nationality) **Movie_Actor**(movie_id, actor_id, role) **Review** (rev_id, user_id, movie_id, rev_text, rating, rev_date) **User** (user_id, username, email, age) i. Find the names of all actors who have acted in at least one movie released after 2015. ii. Find the directors who have directed more than 5 movies. iii. Get the average rating of all movies directed by 'Christopher Nolan'. iv. Find movies that have a higher rating than the average rating of movies released in the same genre. (**solution should be in Nested Query**)	6	L3	3	2

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Unit – II

L3	a)	Consider the two sets of functional dependencies $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ and $G = \{A \rightarrow CD, E \rightarrow AH\}$ Are they equivalent?	4	L3	4	2
	b)	What is Normalization? Explain the first Normal form with example.	2	L2	4	1
4	a)	Suppose a relational schema R (A B C D E F G H I J) and set of functional dependencies X: $\{ \{A,B\} \rightarrow C,\ A \rightarrow \{D,E\},\ B \rightarrow F,\ F \rightarrow \{G,H\},\ D \rightarrow \{I,J\} \}$ What is candidate key of R? Check out whether the relation is in 3NF or not? If not decompose it in 3NF.	4	L3	4	2
	b)	Explain any two Informal Design Guidelines for Relation Schemas.	2	L2	4	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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4th sem MSE II